

Data Understanding

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CS703-152W: Applied Data Science Project

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May 20, 2026

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2.1 Gathering Data

The dataset was obtained from Kaggle. According to the dataset author, the data was scraped from Craigslist in approximately January 2020 and compiled into a single CSV file.

Project scope. This project focuses on rental apartments in the Southern United States (Census Bureau definition: AL, AR, DC, DE, FL, GA, KY, LA, MD, MS, NC, OK, SC, TN, TX, VA, WV). The full dataset spans all 50 states and 12 property types; filtering to the project scope is performed in Phase 3 (Data Preparation). Phase 2 reports below describe the complete collected dataset as obtained from Kaggle.

2.1.1 Data Requirements

The project required a dataset of US rental housing listings containing: the listing price as the target variable, physical attributes (square footage, bedrooms, bathrooms), amenity flags (pets, smoking, wheelchair access, EV charging, furnished), property type, and geographic identifiers (state, region, latitude, longitude). The data needed to include Southern US states and apartments specifically (the project scope), with sufficient volume after filtering (target: ~100K+ records). The selected dataset exceeds these requirements: it covers all 50 states and 12 property types, providing ample records for the Southern-apartment subset.

2.1.2 Selection Criteria and Data Availability

The raw Kaggle file contains 22 columns across 384,977 rows (558.44 MB). Four text/URL columns (id, url, region_url, image_url, description) were excluded from the working dataset, as they carry no predictive signal for price modeling. The remaining 18 columns (four numeric features: price, sqfeet, beds, baths; six binary amenity flags: cats_allowed, dogs_allowed, smoking_allowed, wheelchair_access, electric_vehicle_charge, comes_furnished; four multi-class categoricals: region, type, laundry_options, parking_options; two geographic fields: lat, long; and state) were retained for initial exploration. All further selection decisions are deferred to Task 3.1 (Selecting Data) in Phase 3.

2.1.3 Tool Compatibility

The dataset was loaded into Python using pandas and numpy without any issues. The CSV format was read directly via `pandas.read_csv()` with no encoding errors, parsing failures, or memory constraints (the working file is 37.2 MB). All 18 retained columns were confirmed present and populated upon inspection.

2.1.4 Issues and Difficulties

No difficulties were encountered during data gathering, access, or loading. Initial inspection revealed the presence of extreme outliers (e.g., maximum price over \$2.7B, maximum square footage over 8.3M), \$0 listings, and missing values in several columns; these issues are documented in Task 2.4 (Verifying Data Quality) and will be addressed during Phase 3 (Data Preparation).

2.2 Describing Data

The dataset is a single CSV file (comma-separated values) containing 384,977 rows and 18 columns. It holds a mix of numeric, categorical, text, and geographic feature types. The target variable for this project is price.

The dataset is well-suited for the project's data-mining goals. After filtering to the 17 Southern states and apartments only (Phase 3), approximately 144,000 cleaned records remain across 17 states and several hundred regions: sufficient breadth and volume to train a generalizable price prediction model. All required feature categories (physical attributes, amenities, location, price) are present. Issues such as extreme outliers and missing values exist but are documented in Task 2.4 and will be addressed in Phase 3; they do not prevent proceeding.

2.3 Exploring Data

Exploratory data analysis was conducted using `df.describe()` in Python (pandas) to compute summary statistics for all numeric and binary features. All figures and statistics in this section describe the full 384,977-row collected dataset (all 50 states, all 12 property types); the Southern-apartment subset is analyzed after scope filtering in Phase 3.

Feature Type	Features	Notes
Numeric	price, sqfeet, beds, baths	Target: price. Shows extreme outliers (max \$2.77B, 8.4M sqft). Typical median values: \$1,036, 949 sqft, 2 beds, 1 bath.
Categorical (Binary)	cats_allowed, dogs_allowed, smoking_allowed, wheelchair_access, electric_vehicle_charge, comes_furnished	Boolean amenity flags (0/1). Pets and smoking are common (~72%); wheelchair access (8.2%), EV charging (1.3%), and furnished (4.8%) are minority features.
Categorical (Multi)	region, type, laundry_options, parking_options	Multi-class labels. region: 404 Craigslist metro areas; type: 12 property types (apartment, house, townhouse, etc.); laundry_options: 5 classes; parking_options: 7 classes.
Geographical	lat, long, state	51 states/territories represented; lat/long coordinates cover the continental US, Alaska, and Hawaii. The project scope (Phase 3) filters to the 17 Southern states.

2.3.1 Summary Statistics

The table below presents descriptive statistics for the numeric features and binary amenity flags. Extreme outliers inflate the mean and standard deviation of price and sqfeet, making the median the more representative central-tendency measure.

Feature	Mean	Std	Min	25%	50%	75%
price (\$)	8,826	4.46M	0	805	1,036	1,395
sqfeet	1,060	19,151	0	750	949	1,150
beds	1.91	3.49	0	1	2	2
baths	1.48	0.62	0	1	1	2
lat	37.23	5.55	-43.53	33.45	37.65	41.14
long	-92.70	16.53	-163.89	-100.78	-87.75	-81.18

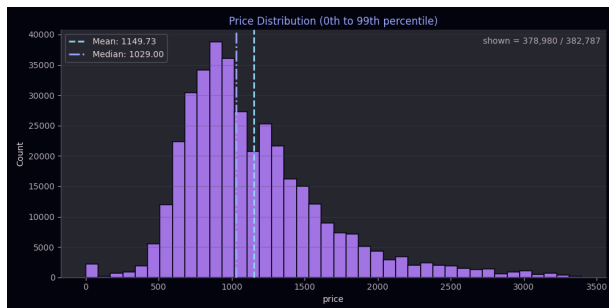
Feature	Mean (Proportion)	Interpretation
cats_allowed	0.727	72.7% allow cats
dogs_allowed	0.708	70.8% allow dogs
smoking_allowed	0.732	73.2% allow smoking
wheelchair_access	0.082	8.2% accessible
electric_vehicle_charge	0.013	1.3% offer EV charging
comes_furnished	0.048	4.8% come furnished

2.3.2 Numeric Distributions

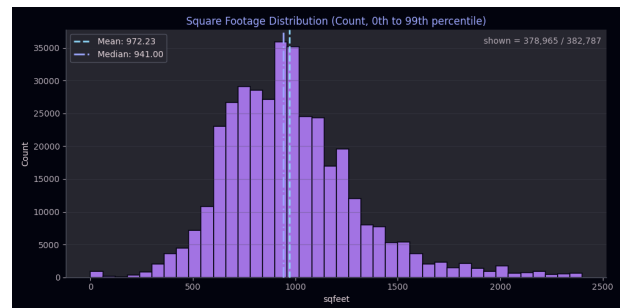
Price and square footage are heavily right-skewed, with the vast majority of listings concentrated below \$2,000 and 1,500 sq ft respectively. Beds and baths show the expected clustering at integer values, with 1–2 bedrooms and 1 bathroom being the most common configurations.

Figure 1

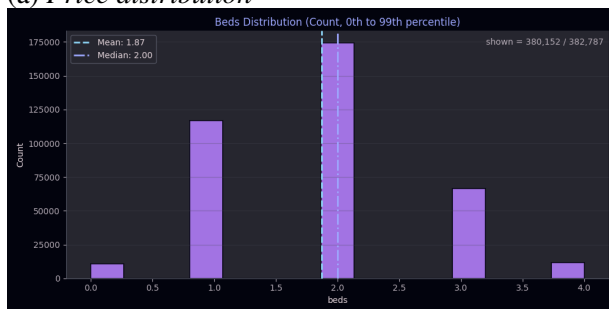
Exploratory data analysis histograms for key housing variables.



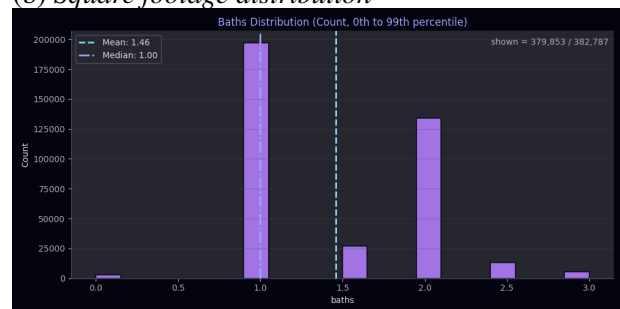
(a) Price distribution



(b) Square footage distribution



(c) Beds distribution



(d) Baths distribution

2.3.3 Categorical Distributions

Pie charts for the six binary amenity flags and two multi-class categoricals are shown in Figure 2. The dataset skews pet- and smoking-friendly ($\sim 70\text{--}73\%$ each), while wheelchair access (8.2%), EV charging (1.3%), and furnished units (4.8%) are minority classes. Laundry and parking options show meaningful class imbalance: the most common laundry arrangement (“w/d in unit”) and parking type (“off-street parking”) each account for a plurality of listings, with “Not Specified” representing a substantial share in both.

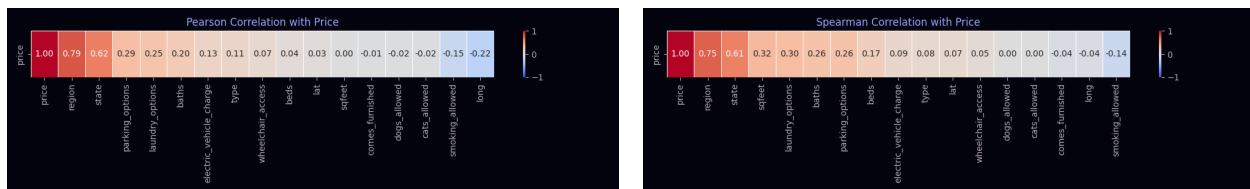
Figure 2
Distribution of selected categorical housing features.



2.3.4 Correlation Analysis

Pearson and Spearman correlation heatmaps (Figure 3) were computed to assess linear and monotonic relationships between numeric features and price. Square footage shows the strongest positive correlation with price (Pearson $r \approx 0.35$, Spearman $\rho \approx 0.55$), while beds and baths have weak-to-moderate positive correlations. Latitude and longitude show negligible linear correlation but a modest Spearman rank correlation, suggesting a non-linear geographic component to pricing.

Figure 3
Correlation Test - Numeric Features vs Price



(a) Pearson

(b) Spearman

2.3.5 Initial Hypotheses

- Square footage will be the single strongest predictor of rental price among the numeric features, consistent with the observed Pearson and Spearman correlations.
- Geographic location (state, region, lat/long) will capture substantial price variation not explained by physical attributes alone, as rental markets differ across Southern metro areas (e.g., Atlanta vs. rural Mississippi) despite regional homogeneity relative to the full US.
- Minority amenity flags (wheelchair access, EV charging, furnished) may have limited predictive power due to severe class imbalance, but their presence may signal premium pricing where they do occur.

2.3.6 Data Quality Signs

Exploration surfaced several data quality concerns that warrant formal verification in Task 2.4:

- Extreme outliers: price reaches \$2.77B and sqfeet reaches 8.39M, orders of magnitude above the 75th percentile. These distort the mean and inflate variance.

- Zero-valued listings: 1,307 records have price = \$0 and 48 have sqfeet < 1. These are unlikely to represent genuine rental transactions.
- Missing values: laundry_options and parking_options are visibly incomplete in the categorical pie charts (the “Not Specified” slices), and a small number of lat/long coordinates are absent.
- Unrealistic maxima: beds = 1,100 and baths = 75 are implausible for residential rentals and may indicate data-entry errors or non-residential listings.

2.4 Verifying Data Quality

A structured quality assessment was conducted to identify missing values, duplicates, pricing anomalies, and structural issues before data preparation began. Each issue is classified by severity and paired with a proposed remedy.

2.4.1 Missing Values

All 18 columns in the working dataset were checked for missing values. Four columns contain missing data; the remaining 14 columns (id, region, price, type, sqfeet, beds, baths, cats_allowed, dogs_allowed, smoking_allowed, wheelchair_access, electric_vehicle_charge, comes_furnished, state) are complete with 0% missing.

Feature	Missing Count	Missing %	Severity	Proposed Remedy
parking_options	140,687	36.54%	Major	Create “Not Specified” category to preserve all records
laundry_options	79,026	20.53%	Major	Create “Not Specified” category to preserve all records
lat	1,918	0.50%	Minor	Drop rows; geographic imputation is unreliable across 404 regions
long	1,918	0.50%	Minor	Drop rows; same rows as missing lat

2.4.2 Duplicate Rows

A full duplicate-row check was performed across all 18 columns of the working dataset. No duplicate rows were found among the 384,977 records, confirming each listing is unique.

2.4.3 Anomalies and Outliers

- **\$0 listings** (1,307 records, 0.34% of data): These may represent listings open to negotiation but introduce noise into price modeling. **Severity: Minor. Remedy:** Remove these records, as \$0 cannot represent a genuine rental price and would distort regression training.
- **Sub-1 sq ft listings** (48 records, 0.01% of data): Physically impossible as habitable rental space. **Severity: Minor. Remedy:** Remove these records.

- **Extreme price outliers** (94 records above \$100K, plus a maximum of \$2.77B): These fall far outside the interquartile range and inflate the dataset mean to \$8,826 vs. a median of \$1,036. **Severity: Moderate. Remedy:** Apply percentile-based capping or IQR-based filtering during Phase 3 cleaning to retain the bulk of the distribution while removing extreme tails.
- **Extreme square footage outliers** (82 records above 10,000 sq ft, plus a maximum of 8.39M sq ft): Analogous to price extremes. **Severity: Moderate. Remedy:** Same percentile- or IQR-based approach as price outliers.
- **Unrealistic bed/bath counts** (beds ≤ 0 : 10,978 records; baths ≤ 0 : 3,107 records; beds > 20 : 4 records; baths > 10 : 4 records): Bedrooms or bathrooms at 0 are plausible (studio apartments, land rentals), but values exceeding 20 beds or 10 baths are implausible for residential rentals. **Severity: Minor. Remedy:** Retain 0-bedroom and 0-bathroom records (valid studio/land listings); cap or remove records with beds > 20 or baths > 10 .

2.4.4 Overall Quality Verdict

The dataset is of sufficient quality to proceed to Phase 3 (Data Preparation), provided the remedies outlined above are applied. The two most impactful issues are (1) the 36.54% missing rate in `parking_options`, which can be preserved via a “Not Specified” level rather than through deletion, and (2) extreme price/square-footage outliers that would distort model training if left untreated. No data collection rework is required.